

SQL:

This database manages employee details, departments, job roles, locations, salaries, and performance reviews. It connects all these areas to give a full picture of the workforce.

For performance analysis, it shows how employees are rated each year. For salary analysis, it tracks pay and compares it with role-based ranges.

For department growth, it counts employees in each department over time. For strategic insights, it combines all data to guide decisions on hiring, promotions, pay, and planning.

Using database:

```
use company_db;
```

Creating Tables:

Department:

This table stores all the departments in the organization, with each department having a unique ID and name. It helps categorize employees based on their work area, like HR, Finance, or IT.

```
create table Departments1(dept_id int primary key,dept_name varchar(40) not null unique);
```

Location:

This table keeps track of office or branch locations. Each location has a unique ID and name, making it easy to know where employees are based.

```
create table Locations1(loc_id int primary key,loc_name varchar(50) not null);
```

Job Roles:

This table defines different job positions in the company, along with their salary ranges. It ensures clarity about roles like Manager, Developer, or Analyst.

```
create table jobroles(role_id int primary key,role_name varchar(50) not null,min_salary decimal(10,2),max_salary decimal(10,2));
```

Employees:

This is the main table that holds employee details such as name, gender, department, role, location, joining date, and status. It links employees to their department, role, and location.

```
create table employees1(emp_id int primary key,emp_name varchar(50) not null,  
                        gender varchar(40),dept_id int,role_id int,loc_id int,join_date date,  
                        status varchar(20) default 'Active',foreign  
key(dept_id) references Departments1(dept_id),  
                        foreign key(role_id)references jobroles(role_id),  
                        foreign key(loc_id) references Locations1(loc_id));
```

Salary:

This table records salary information for employees, including the amount and the date it became effective. It helps track salary changes over time.

```
create table salaries(salary_id int primary key,emp_id int,salary decimal(10,2),effective_date  
date,foreign key(emp_id) references employees1(emp_id));
```

Performance:

This table stores yearly performance ratings for employees. It helps measure how well each employee is performing, with ratings between 1 and 5.

```
create table performance_review(review_id int primary key,emp_id int,review_year int,rating  
decimal(2,1) check (rating between 1 and 5),foreign key(emp_id) references  
employees1(emp_id));
```

Inserting the values through files:

```
BULK INSERT Locations1  
FROM 'C:\Dataset\Locatioin_dataset - Sheet1.csv'  
WITH(  
FORMAT = 'CSV',  
FIRSTROW = 2,  
FIELDTERMINATOR = ',',
```

```
ROWTERMINATOR = '\n',  
TABLOCK);  
select * from Locations1;
```

```
BULK INSERT Departments1  
FROM 'C:\Dataset\Departments Dataset - Sheet1.csv'  
WITH(  
FORMAT = 'CSV',  
FIRSTROW = 2,  
FIELDTERMINATOR = ',',  
ROWTERMINATOR = '\n',  
TABLOCK);  
select * from Departments1
```

```
BULK INSERT jobroles  
FROM 'C:\Dataset\Job_roles_dataset - Sheet1.csv'  
WITH ( FIRSTROW = 2, FIELDTERMINATOR = ',', ROWTERMINATOR = '\n', TABLOCK );  
select * from jobroles
```

```
BULK INSERT employees1  
FROM 'C:\Dataset\Employee.csv'  
WITH(  
FORMAT = 'CSV',  
FIRSTROW = 2,  
FIELDTERMINATOR = ',',  
ROWTERMINATOR = '\n',  
TABLOCK);  
select * from employees1
```

```
BULK INSERT salaries  
FROM 'C:\Dataset\SALARIES_Data - SALARIES_Data.csv'  
WITH(  
FORMAT = 'CSV',  
FIRSTROW = 2,  
FIELDTERMINATOR = ',',  
ROWTERMINATOR = '\n',  
TABLOCK);
```

```
select * from salaries
```

```
BULK INSERT performance_review  
FROM 'C:\Dataset\performance.xlsx - Sheet1.csv'  
WITH(  
FORMAT = 'CSV',  
FIRSTROW = 2,  
FIELDTERMINATOR = ',',  
ROWTERMINATOR = '\n',  
TABLOCK);  
select * from performance_review
```

Task to perform:

Performance Analysis:

1.Top 10 Performances of employees

```
SELECT TOP 10  
    e.emp_id,  
    e.emp_name,  
    d.dept_name,  
    jr.role_name,  
    AVG(pr.rating) AS avg_rating  
FROM employees1 e  
JOIN Departments1 d ON e.dept_id = d.dept_id  
JOIN jobroles jr ON e.role_id = jr.role_id  
JOIN performance_review pr ON e.emp_id = pr.emp_id  
GROUP BY e.emp_id, e.emp_name, d.dept_name, jr.role_name  
ORDER BY avg_rating DESC;
```

	emp_id	emp_name	dept_name	role_name	avg_rating
1	108	Rohan Gupta	DevOps	SDE 1	5.000000
2	153	Amit Kumar	DevOps	SDE 1	5.000000
3	115	Arijun Iyer	Data	Data Scientist	4.900000
4	171	Sneha Kumar	HR	HR	4.900000
5	196	Rohan Patel	DotNET	Manager	4.900000
6	134	Naveen Rao	DotNET	Data Scientist	4.800000
7	154	Meera Kumar	Data	Manager	4.800000
8	169	Rahul Gupta	DevOps	HR	4.800000
9	192	Divya Rao	Data	Associate	4.800000
10	142	Pooja Rao	HR	HR	4.700000

2.Average rating per department

```

SELECT d.dept_id,
       d.dept_name,
       AVG(pr.rating) AS avg_rating
FROM Departments d
JOIN employees e ON d.dept_id = e.dept_id
JOIN performance_review pr ON e.emp_id = pr.emp_id
GROUP BY d.dept_id, d.dept_name
ORDER BY avg_rating DESC;

```

Results		Messages	
	dept_id	dept_name	avg_rating
1	1	HR	3.094444
2	5	DotNET	2.935000
3	2	Data	2.915789
4	3	DevOps	2.878260
5	4	Java	2.730000

3. Employees eligible for promotions (rating > 4 & 2+ Years)

```
SELECT e.emp_id,  
       e.emp_name,  
       d.dept_name,  
       jr.role_name,  
       AVG(pr.rating) AS avg_rating,  
       DATEDIFF(year, e.join_date, GETDATE()) AS years_in_company  
FROM employees1 e  
JOIN Departments1 d ON e.dept_id = d.dept_id  
JOIN jobroles jr ON e.role_id = jr.role_id  
JOIN performance_review pr ON e.emp_id = pr.emp_id  
WHERE e.status = 'Active'  
GROUP BY e.emp_id, e.emp_name, d.dept_name, jr.role_name, e.join_date  
HAVING AVG(pr.rating) >= 4.0  
       AND DATEDIFF(year, e.join_date, GETDATE()) >= 3  
ORDER BY avg_rating DESC, years_in_company DESC;
```

	emp_id	emp_name	dept_name	role_name	avg_rating	years_in_company
1	153	Amit Kumar	DevOps	SDE 1	5.000000	7
2	108	Rohan Gupta	DevOps	SDE 1	5.000000	4
3	171	Sneha Kumar	HR	HR	4.900000	9
4	169	Rahul Gupta	DevOps	HR	4.800000	9
5	192	Divya Rao	Data	Associate	4.800000	7
6	134	Naveen Rao	DotNET	Data Scientist	4.800000	3
7	117	Kavya Shar...	DevOps	Associate	4.600000	10
8	180	Neha Rao	DotNET	SDE 1	4.500000	11
9	173	Swathi Rao	DevOps	SDE 1	4.500000	9
10	116	Swathi Rao	DotNET	Data Scientist	4.500000	9
11	148	Vikram Nair	Data	SDE 1	4.200000	4
12	106	Neha Gupta	Java	Data Scientist	4.000000	11
13	145	Pooja Gupta	Data	Manager	4.000000	6

4. performance trend over years

```
SELECT  
  review_year,  
  AVG(rating) AS avg_rating,  
  MIN(rating) AS min_rating,  
  MAX(rating) AS max_rating,
```

```

COUNT(*) AS total_reviews
FROM performance_review
GROUP BY review_year
ORDER BY review_year;

```

Results		Messages			
	review_year	avg_rating	min_rating	max_rating	total_reviews
1	2015	2.844444	1.2	4.9	9
2	2016	3.116666	1.3	4.5	6
3	2017	3.566666	3.3	3.8	3
4	2018	3.325000	1.7	5.0	8
5	2019	3.388888	2.3	4.9	9
6	2020	3.100000	1.2	4.9	6
7	2021	2.200000	1.0	3.6	9
8	2022	3.030000	1.6	4.7	10
9	2023	2.921428	1.5	4.8	14
10	2024	2.328571	1.1	4.2	7
11	2025	2.814285	1.1	4.6	7
12	2026	2.741666	1.2	5.0	12

salary analytics

5.Highest and lowest salary per role

```

select max(s.salary),min(s.salary),j.role_id from salaries s inner join employees1 e on
s.emp_id=e.emp_id inner join jobroles j on j.role_id=e.role_id group by j.role_id;

```

	(No column name)	(No column name)	role_id
1	980000.00	550000.00	1
2	935000.00	500000.00	2
3	985000.00	545000.00	3
4	970000.00	505000.00	4
5	995000.00	520000.00	5

6.Salary distribution per department

```

SELECT
    d.dept_id,
    d.dept_name,

```

```

MIN(s.salary) AS min_salary,
MAX(s.salary) AS max_salary,
AVG(s.salary) AS avg_salary,
COUNT(s.salary) AS total_employees
FROM Departments1 d
JOIN employees1 e ON d.dept_id = e.dept_id
JOIN salaries s ON e.emp_id = s.emp_id
GROUP BY d.dept_id, d.dept_name
ORDER BY d.dept_name;

```

	dept_id	dept_name	min_salary	max_salary	avg_salary	total_employees
1	2	Data	515000.00	995000.00	728421.052631	19
2	3	DevOps	500000.00	985000.00	752173.913043	23
3	5	DotNET	560000.00	980000.00	808000.000000	20
4	1	HR	510000.00	920000.00	691111.111111	18
5	4	Java	505000.00	990000.00	750500.000000	20

7.salary vs performance correlation

```

SELECT
    SUM((sp.salary - st.avg_salary) * (sp.rating - st.avg_rating)) /
    SQRT(SUM(POWER(sp.salary - st.avg_salary, 2)) * SUM(POWER(sp.rating - st.avg_rating,
2)))
    AS correlation_coefficient
FROM (
    SELECT
        s.salary,
        p.rating
    FROM employees1 e
    JOIN salaries s ON e.emp_id = s.emp_id
    JOIN performance_review p ON e.emp_id = p.emp_id
) sp
CROSS JOIN (
    SELECT
        AVG(s.salary) AS avg_salary,
        AVG(p.rating) AS avg_rating
    FROM employees1 e
    JOIN salaries s ON e.emp_id = s.emp_id
    JOIN performance_review p ON e.emp_id = p.emp_id

```


) st;

	correlation_coefficient
1	-0.0395042620935628

8. Employees below department average salary

```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    s.salary
FROM employees1 e
JOIN Departments1 d ON e.dept_id = d.dept_id
JOIN salaries s ON e.emp_id = s.emp_id
WHERE s.salary < (
    SELECT AVG(s2.salary)
    FROM employees1 e2
    JOIN salaries s2 ON e2.emp_id = s2.emp_id
    WHERE e2.dept_id = e.dept_id
);
```

	emp_id	emp_name	dept_name	salary
1	103	Kiran Nair	HR	510000.00
2	107	Priya Sharma	HR	530000.00
3	112	Kavya Iyer	HR	555000.00
4	114	Amit Singh	HR	565000.00
5	123	Kavya Reddy	HR	610000.00
6	125	Swathi Sharma	HR	620000.00
7	126	Deepak Sharma	HR	625000.00
8	127	Rohan Reddy	HR	630000.00
9	138	Kavya Gupta	HR	685000.00
10	139	Meera Reddy	HR	690000.00
11	135	Neha Sharma	HR	670000.00

Department Growth

9.hiring trend per year

```
SELECT
    YEAR(join_date) AS hire_year,
    COUNT(*) AS total_hired
FROM employees1
GROUP BY YEAR(join_date)
ORDER BY hire_year;
```

	hire_year	total_hired
1	2015	9
2	2016	9
3	2017	15
4	2018	3
5	2019	9
6	2020	13
7	2021	8
8	2022	8
9	2023	11
10	2024	8
11	2025	7

10.headcount growth by department

```
SELECT
    d.dept_name,
    YEAR(e.join_date) AS hire_year,
    COUNT(e.emp_id) AS headcount
FROM employees1 e
JOIN Departments1 d ON e.dept_id = d.dept_id
GROUP BY d.dept_name, YEAR(e.join_date)
ORDER BY d.dept_name, hire_year;
```

	dept_name	hire_year	headcount
1	Data	2015	2
2	Data	2017	2
3	Data	2019	4
4	Data	2020	1
5	Data	2021	1
6	Data	2022	3
7	Data	2023	3
8	Data	2024	2
9	Data	2025	1
10	DevOps	2015	2
11	DevOps	2016	1
12	DevOps	2017	5

Strategic Insights

12. Employees overdue for promotion

```

SELECT e.emp_id,
       e.emp_name,
       e.join_date,
       DATEDIFF(year, e.join_date, GETDATE()) AS years_in_company,
       AVG(pr.rating) AS avg_rating,
       jr.role_name,
       d.dept_name,
       l.loc_name
FROM employees1 e
JOIN jobroles jr ON e.role_id = jr.role_id
JOIN Departments1 d ON e.dept_id = d.dept_id
JOIN Locations1 l ON e.loc_id = l.loc_id
LEFT JOIN performance_review pr ON e.emp_id = pr.emp_id
GROUP BY e.emp_id, e.emp_name, e.join_date, jr.role_name, d.dept_name, l.loc_name
HAVING DATEDIFF(year, e.join_date, GETDATE()) >= 5
      AND AVG(pr.rating) >= 4.0
ORDER BY years_in_company DESC, avg_rating DESC;

```

	emp_id	emp_name	join_date	years_in_company	avg_rating	role_name	dept_name	loc_name
1	180	Neha Rao	2015-05-18	11	4.500000	SDE 1	DotNET	Hyderabad
2	106	Neha Gupta	2015-10-30	11	4.000000	Data Scientist	Java	Pune
3	117	Kavya Sharma	2016-04-05	10	4.600000	Associate	DevOps	Pune
4	157	Meera Verma	2016-12-28	10	4.300000	SDE 1	Java	Trivandrum
5	171	Sneha Kumar	2017-10-06	9	4.900000	HR	HR	Trivandrum
6	196	Rohan Patel	2017-01-27	9	4.900000	Manager	DotNET	Trivandrum
7	169	Rahul Gupta	2017-12-21	9	4.800000	HR	DevOps	Trivandrum
8	154	Meera Kumar	2017-08-15	9	4.800000	Manager	Data	Hyderabad
9	116	Swathi Rao	2017-05-11	9	4.500000	Data Scientist	DotNET	Pune
10	173	Swathi Rao	2017-12-25	9	4.500000	SDE 1	DevOps	Pune
11	153	Amit Kumar	2019-01-02	7	5.000000	SDE 1	DevOps	Hyderabad

13.High Salary but low performance cases

```

SELECT e.emp_id,
       e.emp_name,
       jr.role_name,
       d.dept_name,
       l.loc_name,
       s.salary,
       pr.rating
FROM employees1 e
JOIN jobroles jr ON e.role_id = jr.role_id
JOIN Departments1 d ON e.dept_id = d.dept_id
JOIN Locations1 l ON e.loc_id = l.loc_id
JOIN salaries s ON e.emp_id = s.emp_id
JOIN performance_review pr ON e.emp_id = pr.emp_id
WHERE s.salary > (
    SELECT AVG(salary)
    FROM salaries
)
AND pr.rating < 3.0
ORDER BY s.salary DESC, pr.rating ASC;

```

	emp_id	emp_name	role_name	dept_name	loc_name	salary	rating
1	199	Rohan Rao	Manager	Java	Trivandrum	990000.00	2.8
2	198	Meera Rao	Associate	DevOps	Pune	985000.00	2.6
3	197	Naveen Gupta	HR	DotNET	Trivandrum	980000.00	1.2
4	195	Swathi Rao	Data Scientist	DevOps	Hyderabad	970000.00	1.6
5	194	Amit Kumar	Data Scientist	DevOps	Hyderabad	965000.00	1.5
6	193	Priya Patel	HR	Java	Hyderabad	960000.00	2.4
7	190	Kavya Iyer	Data Scientist	Data	Trivandrum	945000.00	1.7
8	189	Amit Nair	Associate	DevOps	Pune	940000.00	1.4
9	188	Kiran Iyer	SDE 1	DevOps	Hyderabad	935000.00	1.0
10	184	Pooja Iyer	Manager	Data	Pune	915000.00	2.5
11	179	Sneha Singh	Associate	DotNET	Hyderabad	890000.00	1.8

16.Location-wise salary comparison

```

SELECT l.loc_id,
       l.loc_name,
       COUNT(e.emp_id) AS total_employees,
       ROUND(AVG(s.salary), 2) AS avg_salary,
       MIN(s.salary) AS min_salary,
       MAX(s.salary) AS max_salary
FROM Locations l
JOIN employees1 e ON l.loc_id = e.loc_id
JOIN salaries s ON e.emp_id = s.emp_id
GROUP BY l.loc_id, l.loc_name
ORDER BY avg_salary DESC;

```

	loc_id	loc_name	total_employees	avg_salary	min_salary	max_salary
1	1	Trivandrum	30	772000.000000	500000.00	995000.00
2	2	Hyderabad	36	743194.440000	505000.00	970000.00
3	3	Pune	34	730441.180000	515000.00	985000.00

17. Gender diversity ratio(if added)

```
SELECT d.dept_id,
       d.dept_name,
       COUNT(e.emp_id) AS total_employees,
       SUM(CASE WHEN LOWER(e.gender) = 'male' THEN 1 ELSE 0 END) AS male_count,
       SUM(CASE WHEN LOWER(e.gender) = 'female' THEN 1 ELSE 0 END) AS
female_count,
       ROUND(
         (CAST(SUM(CASE WHEN LOWER(e.gender) = 'female' THEN 1 ELSE 0 END) AS
FLOAT)
         / COUNT(e.emp_id)) * 100, 2
       ) AS female_ratio_pct,
       ROUND(
         (CAST(SUM(CASE WHEN LOWER(e.gender) = 'male' THEN 1 ELSE 0 END) AS
FLOAT)
         / COUNT(e.emp_id)) * 100, 2
       ) AS male_ratio_pct
FROM Departments1 d
JOIN employees1 e ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name
ORDER BY female_ratio_pct DESC;
```

	dept_id	dept_name	total_employees	male_count	female_count	female_ratio_pct	male_ratio_pct
1	1	HR	18	7	11	61.11	38.89
2	5	DotNET	20	8	12	60	40
3	2	Data	19	8	11	57.89	42.11
4	4	Java	20	10	10	50	50
5	3	DevOps	23	15	8	34.78	65.22

18. Median salary calculation

```
SELECT
  AVG(1.0 * salary) AS median_salary
FROM (
  SELECT salary,
         ROW_NUMBER() OVER (ORDER BY salary) AS rn,
         COUNT(*) OVER () AS cnt
  FROM salaries
```

```
) AS t
WHERE rn IN ((cnt + 1) / 2, (cnt + 2) / 2);
```

	median_salary
1	747500.000000

19. Employees with multiple promotions

```
SELECT e.emp_id,
       e.emp_name,
       COUNT(DISTINCT e.role_id) AS role_changes
FROM employees1 e
JOIN salaries s ON e.emp_id = s.emp_id
GROUP BY e.emp_id, e.emp_name
HAVING COUNT(DISTINCT e.role_id) > 1
ORDER BY role_changes DESC;
```

emp_id	emp_name	role_changes
--------	----------	--------------

There is no values to print for the case.

20. Identify leadership pipeline candidates

```
SELECT e.emp_id,
       e.emp_name,
       d.dept_name,
       jr.role_name,
       AVG(pr.rating) AS avg_rating,
       DATEDIFF(year, e.join_date, GETDATE()) AS years_in_company,
       COUNT(DISTINCT e.role_id) AS role_changes
FROM employees1 e
JOIN Departments1 d ON e.dept_id = d.dept_id
JOIN jobroles jr ON e.role_id = jr.role_id
JOIN performance_review pr ON e.emp_id = pr.emp_id
WHERE e.status = 'Active'
GROUP BY e.emp_id, e.emp_name, d.dept_name, jr.role_name, e.join_date
```

HAVING AVG(pr.rating) >= 4.0
 AND DATEDIFF(year, e.join_date, GETDATE()) >= 3
 AND COUNT(DISTINCT e.role_id) >= 1
 ORDER BY avg_rating DESC, years_in_company DESC;

	emp_id	emp_name	dept_name	role_name	avg_rating	years_in_company	role_changes
1	153	Amit Kumar	DevOps	SDE 1	5.000000	7	1
2	108	Rohan G...	DevOps	SDE 1	5.000000	4	1
3	171	Sneha K...	HR	HR	4.900000	9	1
4	169	Rahul Gu...	DevOps	HR	4.800000	9	1
5	192	Divya Rao	Data	Associate	4.800000	7	1
6	134	Naveen ...	DotNET	Data Sci...	4.800000	3	1
7	117	Kavya S...	DevOps	Associate	4.600000	10	1
8	180	Neha Rao	DotNET	SDE 1	4.500000	11	1
9	173	Swathi R...	DevOps	SDE 1	4.500000	9	1
10	116	Swathi R...	DotNET	Data Sci...	4.500000	9	1
11	148	Vikram Nair	Data	SDE 1	4.200000	4	1
12	106	Neha Gu...	Java	Data Sci...	4.000000	11	1
13	145	Pooja Gu...	Data	Manager	4.000000	6	1